

CSI ORIGIN 2026

PROBLEM STATEMENT

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Hackathon / Event Name : CSI ORIGIN 2026

Problem Statement ID : 11

Title : Automated Liquidation Shield & Flash-Repayment Vault

2 BACKGROUND

Current Situation / Context

Overcollateralized lending protocols allow users to borrow assets against deposited collateral. During periods of extreme market volatility, a rapid decline in collateral value can cause a borrower's Health Factor to deteriorate faster than the borrower can respond.

Once the Health Factor falls below the protocol's liquidation threshold, third-party liquidators can repay the borrower's debt and seize a portion of their collateral, typically subject to a liquidation penalty.

This creates a critical timing problem: users can suffer substantial capital loss because they are unable to react within a narrow liquidation window, even when their underlying position could potentially be restored through timely debt repayment or collateral restructuring.

Underlying Problem or Need

The core gap is the absence of an autonomous mechanism capable of continuously monitoring leveraged DeFi positions and proactively restructuring them before liquidation becomes inevitable.

Rather than waiting until a position becomes liquidatable, the system must determine when intervention is necessary, how much intervention is required, and how the position can be restored to a predefined safety threshold while minimising the amount of user capital consumed.

Why This Is Becoming Important

DeFi lending positions can become exposed to rapid changes in collateral value during periods of high market volatility. Network congestion and rapidly changing market conditions can further reduce the time available for manual intervention.

A position that appears safe at one point may approach liquidation within a small number of blocks. This creates a need for automated risk management capable of continuously reassessing positions and responding within the available execution window.

Limitations of the Current Situation

Traditional liquidation protection generally requires borrowers to manually monitor their positions and either deposit additional collateral or repay outstanding debt.

These approaches depend on the borrower being able to identify the risk, determine the appropriate intervention amount, obtain the required liquidity, execute transactions successfully, and respond before liquidation occurs.

During periods of high volatility or network congestion, these requirements can make manual intervention ineffective.

Impact of Leaving This Unresolved

If positions are managed reactively, borrowers may experience forced liquidation and significant collateral losses when their Health Factor falls below the liquidation threshold.

Users may also consume more capital than necessary when attempting to protect positions manually, while poorly timed or incomplete interventions can fail to restore solvency.

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What Is Going Wrong / Missing?

DeFi lending systems lack an autonomous risk-management mechanism capable of dynamically restructuring a leveraged position before it reaches an unrecoverable liquidation state.

The missing capability is not simply an alert when Health Factor becomes low. The system must determine **when intervention should occur, how much debt should be repaid, how much collateral should be released, which collateral should be used, and whether the required transactions remain economically and technically viable.**

Who or What Is Affected?

The problem affects users holding leveraged positions in overcollateralized DeFi lending protocols.

It also affects the underlying lending protocols, collateral markets, decentralised exchanges, flash-loan providers, and other on-chain infrastructure involved in maintaining and restructuring these positions.

Why Is It Difficult or Important?

The system must balance several competing factors simultaneously:

Liquidation Risk ↔ Capital Preservation ↔ Swap Slippage ↔ Flash-Loan Cost ↔ Safety Margin

Intervention that is too late may fail to prevent liquidation, while intervention that is too early or too large may unnecessarily consume user capital.

The system must also determine whether sufficient DEX liquidity exists for the required collateral swap and whether slippage, transaction costs, and flash-loan fees make the intervention economically viable.

Furthermore, the restructuring process must be atomic. If the system cannot complete the required repayment, collateral release, swap, and flash-loan repayment within the same transaction, the operation must not leave the user with an incomplete or worse position.

Consequences

If this problem remains unresolved, users will continue to rely on manual monitoring and reactive intervention during volatile market conditions.

This can lead to avoidable liquidations, unnecessary collateral loss, inefficient use of capital, and failed protection attempts during periods when rapid on-chain execution is critical.

4 CHALLENGE

Design and build an **autonomous liquidation-protection system** capable of continuously monitoring an on-chain lending position, predicting liquidation risk, determining the minimum effective intervention required, and atomically restructuring the position to restore it above a predefined safety threshold.

When intervention is justified, the system should be capable of evaluating and executing a sequence involving temporary liquidity, debt repayment, collateral release, asset swapping, and repayment of the temporary liquidity within a single atomic transaction.

The System Should Operate as a Continuous Protection Loop

Monitor Position → Predict Liquidation Risk → Calculate Optimal Repayment → Source Flash Liquidity → Repay Debt → Unlock Collateral → Swap → Repay Flash Loan → Restore Safe Position

Problem Requirements / Constraints

- The system must continuously monitor relevant lending-position parameters and identify when the position approaches a predefined liquidation-risk threshold.
- The system must determine how much debt needs to be repaid to restore the position to a predefined safety threshold while minimising user capital consumption.
- The system must determine the appropriate amount and source of collateral to be released or converted to support the repayment.
- The system must evaluate available DEX liquidity, expected slippage, flash-loan fees, and transaction costs before determining whether an intervention is economically viable.
- The system must support atomic execution of the required restructuring sequence, including temporary liquidity, debt repayment, collateral release, asset conversion, and repayment of the temporary liquidity.
- The system must ensure that an intervention does not leave the user with an incomplete or materially worse position if the required atomic conditions cannot be satisfied.
- The system must dynamically determine an appropriate safety buffer based on relevant risk and market conditions rather than relying exclusively on a fixed liquidation threshold.
- The system must act proactively before the position reaches the protocol's liquidation threshold wherever sufficient information and execution conditions allow.

5 ANNEXURE — DETAILED RULES / CONSTRAINTS

Health Factor Monitoring

The system must continuously evaluate the position's Health Factor and relevant collateral and debt conditions. Risk assessment should occur before liquidation becomes inevitable rather than relying solely on the protocol's final liquidation threshold.

Intervention Sizing

The system should determine the minimum effective debt repayment or position restructuring required to restore the desired safety level. Interventions should avoid consuming unnecessary user capital.

Collateral Selection

Where multiple collateral assets are available, the system should evaluate which collateral can be most appropriately used for restructuring based on factors such as value, liquidity, swap conditions, and resulting position risk.

Flash Liquidity

Temporary liquidity may be used to facilitate atomic position restructuring. The system must account for the cost and availability of the required flash liquidity when evaluating whether an intervention is viable.

DEX Liquidity and Slippage

The system must assess whether sufficient market liquidity exists to convert released collateral into the required repayment asset. Expected slippage and associated execution costs must be considered before execution.

Atomic Settlement

The complete protection operation must succeed atomically. If the required conditions cannot be satisfied within the same transaction, the transaction should revert rather than leaving the user with an incomplete restructuring or unrepaid temporary liquidity.

Safety Margin

The system should target a safety threshold above the protocol's liquidation boundary. The appropriate margin should account for relevant market volatility and the possibility of continued adverse movement.

Economic Viability

The system must consider whether the expected cost of intervention, including flash-loan fees, swap costs, slippage, and transaction costs, is justified relative to the liquidation risk and capital being protected.

Autonomous Execution

The system should minimise the need for continuous human intervention. Human-defined risk preferences or safety parameters may govern the system, but the agent should independently determine when and how an eligible intervention should occur.

6 FINAL PROBLEM CONSTRAINTS

- The system must proactively identify and manage liquidation risk rather than merely notify the user after the position becomes liquidatable.
- The system must determine the **minimum effective intervention** required to restore the position above a predefined safety threshold.
- The system must account for liquidation risk, collateral value, available liquidity, swap slippage, flash-loan fees, transaction costs, and market volatility.
- The complete restructuring operation must execute **atomically**, with failure resulting in transaction reversion rather than an incomplete or worse position.
- The system must avoid unnecessary consumption of user capital when determining the intervention amount.
- The system must account for the possibility of continued adverse market movement by maintaining an appropriate safety buffer above the liquidation threshold.
- The system must ensure that the protection mechanism itself does not introduce additional insolvency, excessive leverage, or economically unjustified capital loss.
- The challenge is focused on **autonomous on-chain risk management and atomic position restructuring**, not merely a liquidation alert system or a conventional automated repayment bot.